About Besmile

Besmile is a global digital solutions provider for restorative and implant dentistry.

We offer integrated systems—premium CAD/CAM materials, advanced equipment, and precision implant systems—designed to streamline workflows and ensure reliable results.

All core products are developed and manufactured in-house, ensuring consistent quality and continuous innovation.

Trusted by over 1,000 partners in more than 100 countries, we empower dental professionals to create confident, lasting smiles.

Technology creates the best smile.

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3D Aconia® Ultra

Aesthetics,
Stronger than Ever



3D Aconia® Ultra

Aesthetics , Stronger than Ever

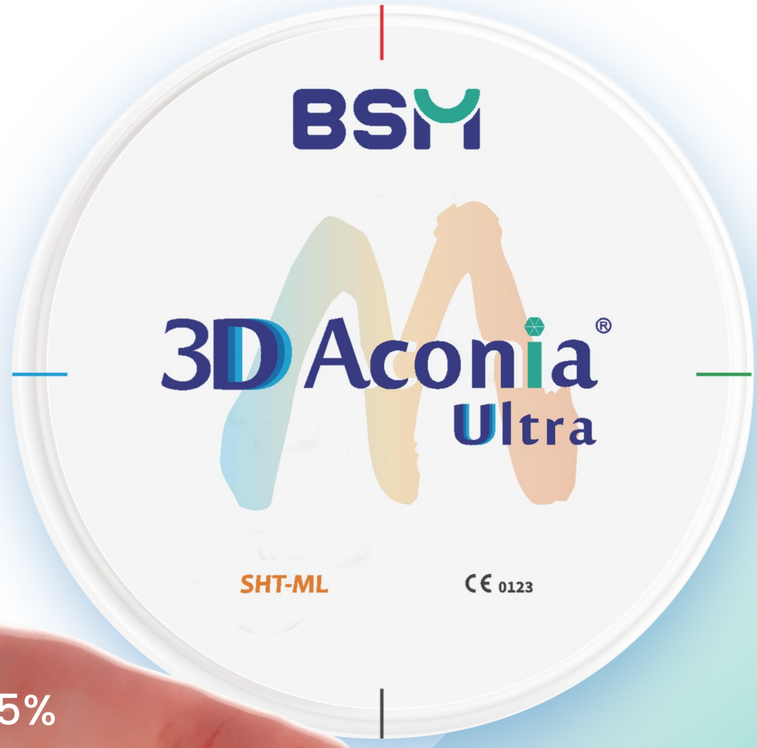
3D Aconia Ultra unites exceptional **aesthetics** and durable **performance** in one premium material.

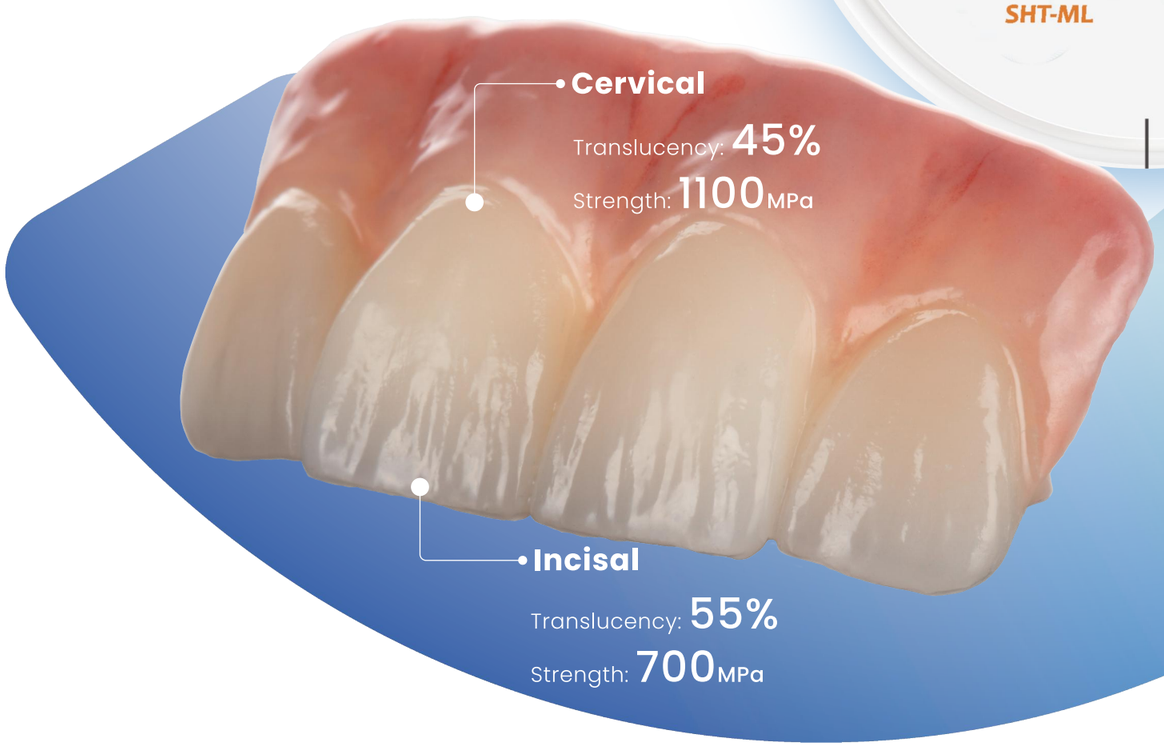
Its **ultra-translucent** multilayer design delivers lifelike gradients, reliable strength, and predicatable results —a standout choice for consistently beautiful restorations.

Aesthetics

Strength

Durability





Superior Translucency

Ultra-high translucency and seamless gradient deliver realistic dimension and lifelike restorations.

Flexible Sintering, Consistent Results

Within a flexible 1520°C–1560°C sintering range, providing stable, predictable shade and translucency every time.



Glaze n' Go!

No stain needed, provides accurate shading and seamless 3D effect, delivers ideal, ready-to-go restorations immediately.



No More Rework

Low deformation and high fracture toughness (≥5 MPa·m$^{1/2}$) ensure accurate, durable bridges with fewer fractures and far less rework.



3D Multilayer Technology

PATENTED TECHNOLOGY · U.S. PATENT

10 yrs of R&D

10,000+ Tests

U.S. Patented

Challenges You Often Face

- Too many SKUs. Too much inventory.
- Visible shade steps. Uneven layering.
- Deformation. Unstable results.

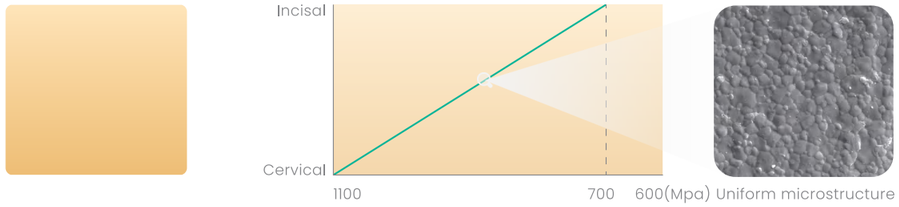
How 3D Aconia Ultra Helps

- One material. Many indications.
- A seamless gradient. Peak aesthetics.
- Near-zero deformation. Consistent performance.

How it works

Interleaved Layer Structure

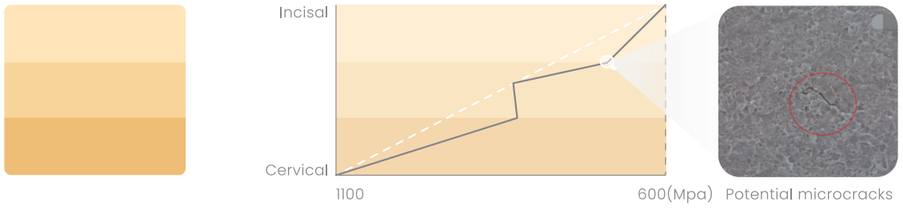
3D Aconia Ultra Multilayer



✓ Seamless gradient

✓ Uniform strength distribution

Conventional Structure



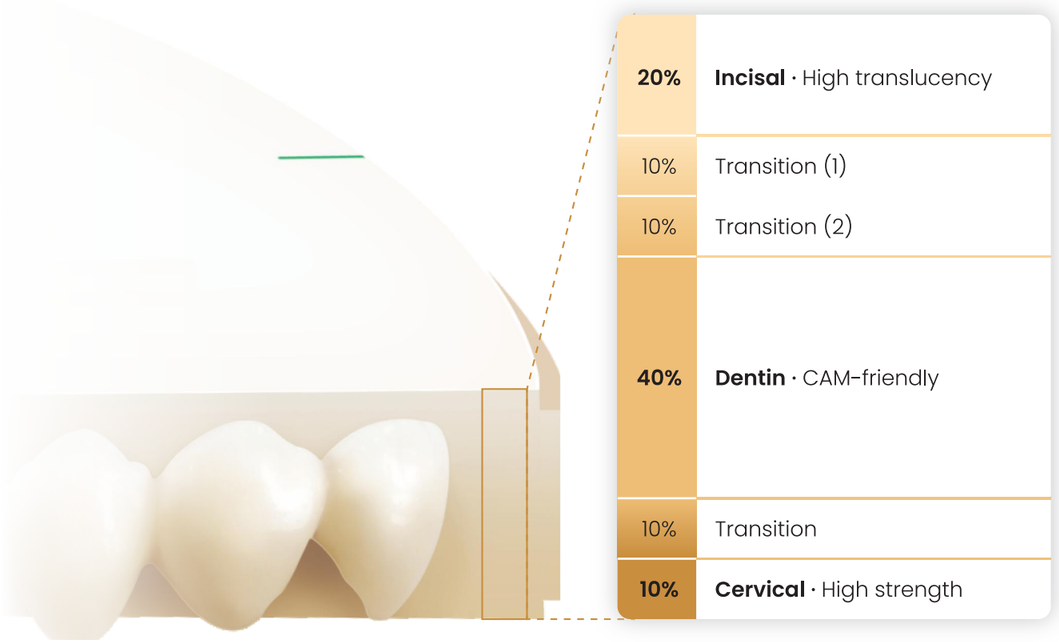
✗ Uneven color transition

✗ Strength decreases at the interlayer zone

Our patented interleaved technology upgrades conventional multilayer zirconia, significantly enhancing interlayer bonding and shear resistance.

Optimized powder formulations and precise blending ratios deliver higher strength, better translucency, and more uniform shrinkage.

Look into Details



Indications



Inlays/Onlays · Copings · Full contour crowns · Veneer · Full contour bridge (⩽7 units) · Full contour bridges (⩽14 units) · Implant-supported restorations

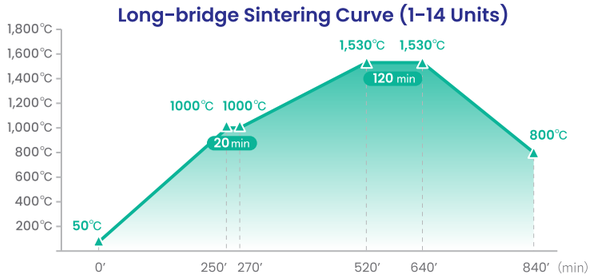
Technical Parameters

Flexural strength (3-point)	700-1100Mpa
Translucency	45%-55%
Shade	A1, A2, A3, A3.5, A4, B1, B2, B3, B4, C1, C2, C3, C4, D2, D3, D4, 0M1, 0M2, 0M3
Disc thickness (ϕ98mm)	12/14/16/18/20/22/25/28*/30*
Sintered density	＞6.02
Vickers-hardness HV10	1300±50
Fracture toughness	＞5

**Customized production required. Please consult your local sales representative for details.*

Sintering Curve

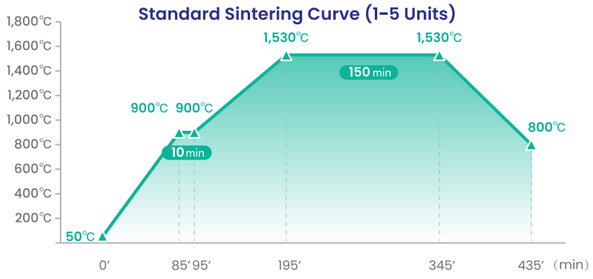
Long-bridge Sintering Curve (1-14 Units)



Long-Bridge Sintering Parameters (1-14 Units)

Step	Initial temperature(℃)	Final temperature(℃)	Time(min)	Heating rate(℃/min)
1	50	1000	250	4
2	1000	1000	20	Holding
3	1000	1530	250	2
4	1530	1530	120	Holding
5	1530	800	200	-4
6	800	Natural cooling		

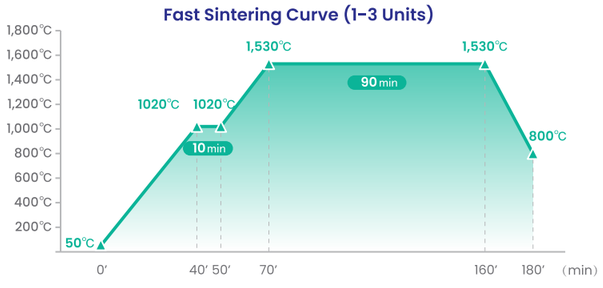
Standard Sintering Curve (1-5 Units)



Standard Sintering Parameters (1-5 Units)

Step	Initial temperature(℃)	Final temperature(℃)	Time(min)	Heating rate(℃/min)
1	50	900	85	10
2	900	900	10	Holding
3	900	1530	100	6
4	1530	1530	150	Holding
5	1530	800	90	-8
6	800	Natural cooling		

Fast Sintering Curve (1-3 Units)



Fast Sintering Parameters (1-3 Units)

Step	Initial temperature(℃)	Final temperature(℃)	Time(min)	Heating rate(℃/min)
1	50	1020	40	24
2	1020	1020	10	Holding
3	1020	1530	20	25
4	1530	1530	90	Holding
5	1530	800	20	-36
6	800	Natural cooling		

*Best results are achieved when used with the BSM FM30 furnace in Standard Mode.

Digital Workflow

Unparalleled Efficiency,Unimaginable Ease

Scanning

The BSM intraoral scanner M5 pro collects visualized and simultaneous information from the patients within 10 minutes.



Designing

The Smile Cloud design platform (https://www.bsmsmilecloud.com/) will provide the design service you need with fascinating design results.



Selecting Materials

3D Aconia Ultra delivers superior performance with excellence in aesthetics and function.



Milling

The BSM-500D is a 5-axis dry milling machine that produces high-precision zirconia restorations with 5μm milling accuracy.



Sintering

The BSM-FC40 zirconia sintering furnace offers fast sintering with a capacity of up to 80 crowns sintered at once.



Finishing

Artamic Stain/Glaze offers a wide range of shades for customization and characterization, delivering aesthetics and quality in restorations.

